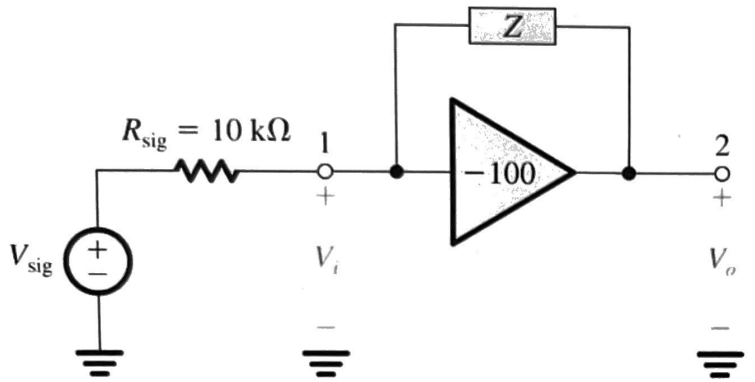
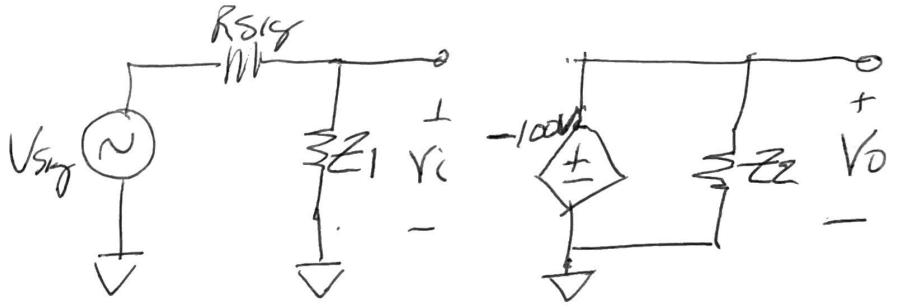
Example 10.4

Find the Miller equivalent circuit when Z is a 1 MΩ resistor and determine V_o/V_{sig}





$$Z_1 = \frac{Z}{1-K} = \frac{1M\Omega}{1-(-100)} = \frac{1M\Omega}{101} = \boxed{9.9k\Omega}$$

$$Z_2 = \frac{Z}{1-\frac{1}{K}} = \frac{1M\Omega}{1+\frac{1}{100}} = \boxed{0.99M\Omega}$$

$$\frac{V_o}{V_{sig}} = \frac{V_o}{V_i} \times \frac{V_i}{V_{sig}}$$

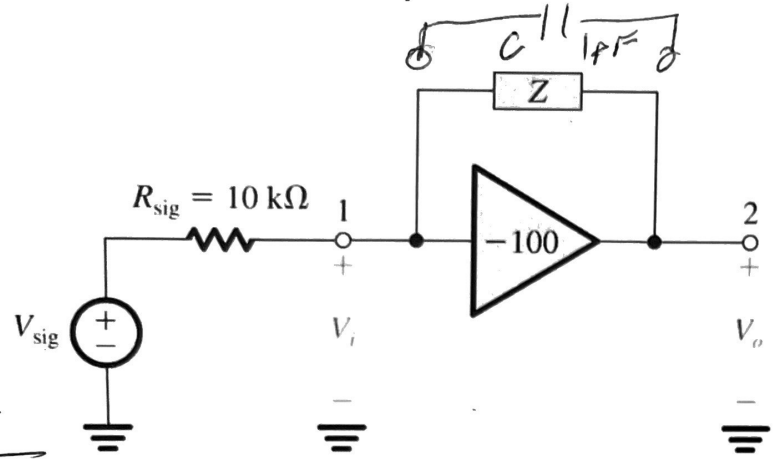
$$= -100 \times \frac{Z_1}{Z_1 + R_{sig}}$$

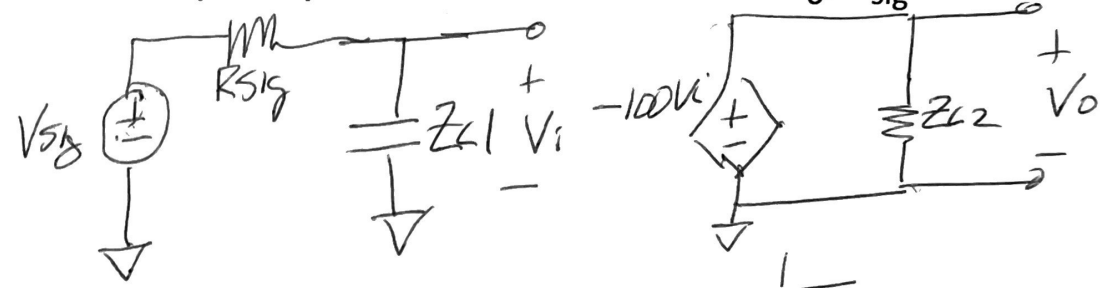
$$= (-100)\left(\frac{9.9}{9.9+10}\right)$$

$$\boxed{\frac{V_o}{V_{sig}} = -49.7 \frac{V}{V}}$$

Example 10.4

Find the Miller equivalent circuit when Z is a 1 pF capacitor and determine V_o/V_{sig}





$$Z_{C1} = \frac{Z}{1-K}$$

$$Z_{C1} = \frac{\frac{1}{sC}}{1+100} = \frac{1}{s(101C)} \therefore \boxed{C_1 = 101\,pF}$$

$$Z_{C2} = \frac{Z}{1-\frac{1}{K}} = \frac{\frac{1}{sC}}{\left(1+\frac{1}{100}\right)} = \frac{1}{sC(1.01)} \Rightarrow \boxed{C_2 = 1.01\,pF}$$

$$\frac{V_o}{V_{sig}} = \frac{V_o}{V_i} \times \frac{V_i}{V_{sig}} = (-100)\frac{\frac{1}{sC_1}}{\left(\frac{1}{sC_1}+R_{sig}\right)}$$

$$= \frac{-100}{1+sC_1R_{sig}}$$

$$= \frac{-100}{1+s(101\,pF)(10\,k\Omega)}$$

$$\boxed{\frac{V_o}{V_{sig}} = \frac{-100}{1+\underset{j\omega}{s}\underbrace{(1.01\times10^{-6})}_{\tau}}}$$

$\omega_H = 990.1$ Krad/s

$\boxed{f_H = 157.6\ KHz}$ −3dB point